

Integrating Analytic Hierarchy Process (AHP) and Bayesian Best-Worst Method (BWM): A dual-phase MCDM framework for addressing critical challenges in lean manufacturing systems

Samieyul Islam Yamaan^{1*}, Shamsul Arafat Chowdhury², Ashrafe Bin Alim³,
Md. Limonur Rahman Lingkon⁴, Riasad Rahman Rakib⁵, Md. Abdullah Al
Kafi⁶ and Md. Imtiaz Kabir Niloy⁷

Department of Industrial & Production Engineering, Rajshahi University of Engineering & Technology, Rajshahi-6204, Bangladesh.

*Email: yamaan2105030@gmail.com¹

Abstract: Lean manufacturing has become a vital part of operational excellence. In a world of constant change, lean manufacturing provides the crucial framework for maximizing value and minimizing waste. However, critical problems and selecting optimal solutions in complex production systems remain challenging. This study proposes a novel dual-phase multi-criteria decision-making (MCDM) framework to the system to address these gaps systematically. In the first phase, the Analytic Hierarchy Process (AHP) is employed to identify the most impactful and dangerous problems hindering lean manufacturing efficiency, evaluating criteria such as impact on efficiency, waste reduction, cost saving, scalability of improvement, alignment with lean, etc. The second phase introduces the Bayesian Best-Worst Method (Bayesian-BWM) to prioritize context-specific solutions for the most critical problem, incorporating probabilistic reasoning to enhance decisions under uncertainty. Experts from various industries and the manufacturing unit validate the framework. The results reveal the best solution among the alternatives and compare them with each other with the help of the necessary software. The Bayesian Best-Worst method is a powerful decision-making tool for finding the best solution alongside essential coding. The hybrid AHP-Bayesian BWM approach outperformed standalone MCDM methods by reducing subjective biases in problem identification and improving solution reliability through Bayesian posterior probabilities. This research is all about making lean manufacturing work. Companies often struggle to identify a problem and find the right lean solution. Therefore, a structured, data-driven approach is developed to help them make those tough decisions for continuous improvement. Combining the analytical AHP method with the probabilistic Bayesian-BWM can effectively tackle complex industrial challenges, providing a comprehensive, reliable, rigorously tested and validated solution.

Keywords: Lean Manufacturing, Analytic Hierarchy Process (AHP), Bayesian, multi-criteria decision-making (MCDM).

1. Introduction

Lean Manufacturing (LM) has emerged as a widely adopted methodology for enhancing operational efficiency, minimising waste, and improving overall productivity in manufacturing systems [30]. The birthplace of modern lean is arguably Toyota in post-World War II Japan. Facing limited resources and a need to be competitive, they developed the Toyota Production System (TPS). This system focused on eliminating waste, continuous improvement (kaizen), and respecting workers. However, the successful implementation of LM is often hindered by various critical challenges, such as overproduction inventory excess, motion waste, defects, and waiting time [31]. These challenges can be overcome only through a structured decision-making process that considers several factors, optimises significant ones and provides the best solution for a specific problem.

Multi-criteria decision-making (MCDM) methods provide a systematic approach to tackling complex decision problems in LM. The Analytic Hierarchy Process (AHP) and the Bayesian Best-Worst Method (BWM) have gained significant attention for their effectiveness in handling qualitative and quantitative data [32]. AHP is a well-known hierarchical decision-making method that relies on pairwise comparisons to establish priorities after arranging them in a hierarchy. Meanwhile, Bayesian BWM employs probabilistic reasoning to improve decision-making under uncertainty. It requires fewer pairwise comparisons than AHP. So, it is less time-consuming.

As existing studies had gaps in integrating the two methods, this paper presents a practical two-stage MCDM model based on AHP and Bayesian BWM to address critical issues in LM systems. The proposed model leverages the strengths of both methods: the structured ranking of AHP and the superiority of Bayesian BWM in managing uncertainty. Through this mathematical framework, organizations can make more logical, data-driven, and evidence-based decisions to optimize the efficiency and effectiveness of their lean manufacturing processes. The framework's application in a real industrial manufacturing setting, supported by the necessary software, demonstrates its potential to transform decision-making in lean manufacturing environments, providing a practical and reliable solution for the challenges faced in lean manufacturing.

Necessary steps have been listed for implementing feasible solutions towards critical problems. The rest of this paper is structured as follows: The post-introduction section offers a literature review of the challenges with LM and MCDM approaches. Section 3 describes the proposed dual-phase methodology, effectively explaining the AHP and Bayesian BWM combination. The following section examines the application and verification of the method using the necessary software in an actual industrial manufacturing setting. The last section presents conclusions and offers potential future research avenues. References are given as per rule.

2. Literature Review

Lean manufacturing (LM) is intended to enhance product quality and process capability while eliminating all forms of waste from a manufacturing process. A survey of 127 manufacturing companies showed that underdeveloped countries still have difficulty implementing LM. However, it has been discovered that although steps have already been taken to deploy LM, progress is being made slowly [1]. Implementing lean manufacturing depends on success factors to make it easier to implement. Still, obstacles, including overproduction, inventory excess, defects, motion waste, and waiting time, are found to be challenging to adopt and maintain, and sometimes failure even occurs. Research has highlighted a structured framework that aligns with the actual lean practice applied while identifying the key factors driving the adoption of lean manufacturing in the Indian sector [2]. Five essential success factors are needed for successful

deployment [3]. These are identified as Inventory Reduction, Production Flexibility, Waste Reduction, Cost Reduction, and Process Optimization. At the same time, high levels of institutional collectivism, future orientation, humane orientation, and low assertiveness are exhibited by successful lean plants [4]. The research also extensively employed soft lean practices like small group problem solving, employee training, supplier partnerships, customer involvement, and continuous improvement. In this paper, these factors are addressed through Just-In-Time (JIT) Production, Kanban System, Demand Forecasting and Planning, Reduced Batch Sizes, and Continuous Improvement (Kaizen). In the past, many papers have been published to identify the critical factors of successful lean implementation using different MCDM methods, such as identifying and ranking essential success factors (CSFs) for sustainable lean manufacturing implementation, which are ranked using the Analytical Hierarchy Process [5]. However, no connection has been made to essential problems caused by the failure of lean implementation to determine the optimal solution. At the same time, CSFs for sustainable lean manufacturing in the Bangladeshi furniture industry have been identified and grouped under organisational support, supply chain management, and performance/technology. The Bayesian Best- Worst Method has evaluated key priorities such as technology adoption, managerial support, lead time reduction, and resource efficiency. Still, standards have not been specified, and a solution for the significant issues with lean implementation has not been provided [6].

Despite the enormous advantages of LM deployment that have been emphasised, few companies have been found to implement this system successfully. Numerous problems and issues have been reported regarding the failure of LM implementation. A study has identified barriers to successful lean implementation. However, information about the significant factors of failure and the solutions has been neglected [7].

A model has described successful lean application as the company-specific application of lean principles rather than a one-size-fits-all approach [8]. Most companies have been found to fail due to a project-based mindset rather than embracing lean tools into daily routines. Many studies have shown that there are many obstacles to LM deployment. It was concluded by a study that employee resistance, high managerial pressure, and inadequate communication were significant barriers in Polish businesses [9]. Tools such as 5S, Kaizen, Kanban, Poka-Yoke, and TPM eliminate several types of waste. Every lean tool exhibits different levels of effectiveness for certain types of garbage. However, these papers have not concentrated on the most significant and hazardous issue impeding lean manufacturing efficiency and the best way to address it.

According to the literature analysis, inventory reduction, production flexibility, waste reduction, cost reduction, and process optimisation are the primary requirements for lean manufacturing. According to the barriers and lean waste-related papers, overproduction, excess inventory, defects, motion waste, and waiting time have been identified as the main issues for lean implementation. Just-In-Time (JIT) production, the Kanban system, demand forecasting and planning, reducing batch sizes, and continuous improvement (kaizen) have been proposed as the solutions. These criteria, critical problems, and solutions have been examined from every aspect of each article, with the most prevalent and significant ones being highlighted.

2.1 Research Gap

Many studies have been conducted on the obstacles to successful lean manufacturing implementations and the failures of lean manufacturing using expert opinions and MCDM techniques like the Bayesian BWM method or the analytical hierarchy process. However, each study has a different concept, and the most optimal solution for implementation and failure is missing. However, there is a lack of combining the MCDM method for examining lean effectiveness

and what the primary issue that leads to successful or unsuccessful lean practice. Additionally, no relationship has been established to identify or select the best problem to erase.

3. Methodology

This research employs a two-phase methodology integrating the Analytic Hierarchy Process (AHP) and Bayesian Best-Worst Method (BWM) to address lean manufacturing challenges and solutions. In Phase 1, the work was conducted using Excel, five criteria (e.g., impact on efficiency, waste reduction, cost saving, implementation difficulty, alignment with lean) and five critical problems including overproduction, waiting time were defined through literature and expert input. Industry experts performed pairwise comparisons via a 9-point Saaty scale, with Excel automating weight calculations (via matrix functions like MMULT), consistency checks ($CR < 0.1$), and ranking of problems based on global priority scores. After arranging them in a table, the normalisation process was done. After the proper summation process, the weight has been calculated perfectly. The process was repeated for the five alternatives. After calculating the weights of five options, an appropriate multiplication process has been done to allocate the final weights of those alternatives based on the five described criteria. Phase 2 (Bayesian BWM) was implemented in MATLAB, where solutions for the top-ranked problem were prioritized using probabilistic reasoning. Experts provided "Best" and "worst" judgments for five solutions, including the Kanban system, just-in-time production, and demand forecasting and planning. MATLAB processed these inputs to estimate posterior weights, credibility intervals, and solution rankings. In the "Best to Others" approach, each criterion in the row was compared to the others, with numbers assigned to represent the preference among the available options. In contrast, the reverse was done for the "Worst to Others" approach. The hybrid framework was validated through workshops and a real-world case study, showcasing reduced subjective bias (AHP) and enhanced reliability under uncertainty (Bayesian BWM). Excel was utilized due to its accessibility, enhancing interactivity among all users. MATLAB's powerful toolbox was used for statistical and probabilistic analysis. The Bayesian Best-Worst Method employed the Monte Carlo simulation, specifically, Markov Chain Monte Carlo (MCMC), to estimate the posterior distribution of the weight vector. The algorithm was implemented in MATLAB, and the JAGS tool was integrated for Bayesian inference. Thus, a practical, data-driven method was developed to help companies transition from problem identification to selecting the best lean solutions. It was designed to be scalable for operations of any size.

4. Result Analysis

The study utilised a two-phase multi-criteria decision-making framework. In Phase 1, the AHP was applied to identify the most critical problem in lean manufacturing. Five criteria—Impact on Efficiency (C1), Waste Reduction (C2), Cost Saving (C3), Implementation Difficulty (C4), and Alignment with Lean (C5)—were evaluated using pairwise comparisons by industry experts on a 9-point Saaty scale. The pairwise comparison matrix of criteria is created in Table 1 to assess each criterion's relative value to the others. Impact on Efficiency, Waste Reduction, Cost Saving, Scalability of Improvement, and Lean Alignment are among the parameters considered. Values are assigned according to the relative importance of each criterion after pairwise comparisons. For instance, Impact on Efficiency is three times more important than Waste Reduction, five times more important than Cost Saving, four times more important than Scalability of Improvement, and two times more important than Alignment with Lean. It is also given a higher priority (value = 1) when compared to itself. The weights assigned to each criterion in the decision-making process are established using the matrix. The pairwise comparison matrix is normalized to ensure

that the sum of all the columns in Table 2 equals 1. The relative weights of each criterion are determined by this normalizing method. Each element in the pairwise comparison matrix is divided by the sum of its corresponding column to obtain the normalized values. For instance, by dividing its initial value (1) by the total of the columns (2.28571428), the normalized value for Impact on Efficiency is determined to be 0.43795620. The final weights of the criteria, crucial for ranking and decision-making, are calculated using the normalized matrix. The eigenvalue method in Excel computed normalised criteria weights (e.g., C1: 41.8%, C2: 16%, C3: 9.4%) and consistency ratios (CR < 0.1).

Table 1. Pairwise Comparison Matrix of Criteria

Criteria	Impact on Efficiency	waste reduction	cost saving	scalability of improvement	Alignment with Lean
Impact on Efficiency	1	3	5	4	2
waste reduction	0.33333333	1	2	3	0.5
cost saving	0.2	0.5	1	2	0.33
Implementation Difficulty	0.25	0.33333333	0.5	1	0.25
Alignment with Lean	0.5	2	3	4	1

Five alternatives—Overproduction, Inventory Excess, Defects, Motion Waste, and Waiting Time—were then ranked against these criteria.

Table 2. Normalised AHP Criteria Pairwise Comparison Matrix

Criteria	Impact on Efficiency	waste reduction	cost saving	scalability of improvement	Alignment with Lean	weights
Impact on Efficiency	0.43795620 4	0.43902439	0.43478260 9	0.28571428 6	0.48979591 8	0.41745468 1
waste reduction	0.14598540 1	0.14634146 3	0.17391304 3	0.21428571 4	0.12244898	0.16059492
cost saving	0.08759124 1	0.07317073 2	0.08695652 2	0.14285714 3	0.08163265 3	0.09444165 8
scalability of improvement	0.10948905 1	0.04878048 8	0.04347826 1	0.07142857 1	0.06122449	0.06688017 2
Alignment with Lean	0.21897810 2	0.29268292 7	0.26086956 5	0.28571428 6	0.24489795 9	0.26062856 8

The exact process has been followed for those alternatives. Global weights were derived by multiplying local alternative weights with criteria weights, yielding Overproduction as the top problem (weight = 2.087), followed by Waiting Time (1.312). In Phase 2, the Bayesian Best-Worst Method (BWM) prioritised solutions for Overproduction. Experts rated Just-in-Time Production as the “best” solution and Takt Time Optimization as the “worst.” A probabilistic model in MATLAB employed Markov Chain Monte Carlo (MCMC) sampling to estimate posterior weights, with Just-in-Time Production emerging as the optimal solution (weight = 0.3173), supported by narrow credibility intervals (95% CI: 0.29–0.34), confirming statistical significance over alternatives like Pull-Based Scheduling (0.256). Tables 1 to 5 are used to evaluate the criteria, and Figure 1 represents the Criteria Comparison Network for the Bayesian Best-Worst Method (BWM).

Table 3. Pairwise Comparison Matrix of Criteria

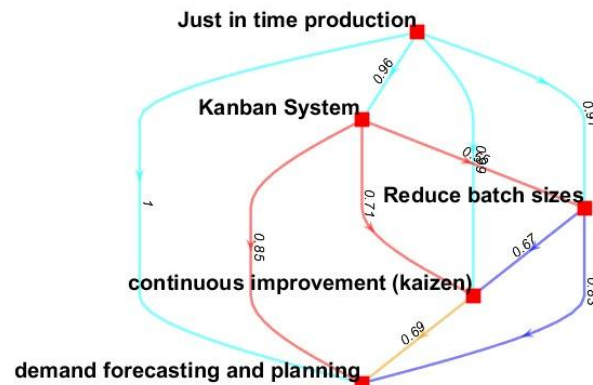
Alternative	Overproduction	Inventory excess	Defects	Motion Waste	waiting time
Overproduction	1	5	3	4	2
Inventory excess	0.2	1	0.30	0.5	0.25
Defects	0.333333333	3	1	2	0.50
Motion Waste	0.25	2	0.5	1	0.30
waiting time	0.5	4	2	3	1

Table 4. Normalised AHP Alternatives Pairwise Comparison Matrix

Alternative	Overproduction	Inventory excess	Defects	Motion Waste	waiting time	weights
Overproduction	0.437956204	0.333333333 33	0.4411764 71	0.3809523 81	0.4938271 6	2.0872455 5
Inventory excess	0.087591241	0.066666666 67	0.0441176 47	0.0476190 48	0.0617283 95	0.3077229 97
Defects	0.109489051	0.133333333 33	0.0735294 12	0.0952380 95	0.0740740 74	0.4856639 66
Motion Waste	0.109489051	0.133333333 33	0.0735294 12	0.0952380 95	0.0740740 74	0.4856639 66
waiting time	0.218978102	0.266666666 67	0.2941176 47	0.2857142 86	0.2469135 8	1.3123902 82

Table 5. Alternative Evaluation Matrix

Alternative	Impact on Efficiency	waste reduction	cost saving	scalability of improvement	Alignment with Lean	Total score
Overproduction	0.871330426	0.335201033	0.19712293	0.139595342	0.543995818	2.08724555
Inventory Excess	0.128460406	0.04941875	0.02906187	0.020580567	0.080201404	0.307722997
Defects	0.202742696	0.077995166	0.04586691	0.03248129	0.126577904	0.485663966
Motion Waste	0.202742696	0.077995166	0.04586691	0.03248129	0.126577904	0.485663966
waiting time	0.547863467	0.210763213	0.123944314	0.087772888	0.3420464	1.312390282

**Figure 1.** Bayesian Best-Worst Method (BWM) Criteria Comparison Network

5. Discussion

The results underscore Overproduction's dominance as a critical lean manufacturing challenge, aligning with its systemic impact on efficiency (C1) and resource waste (C2). Its weight (2.087) reflects its role in triggering downstream inefficiencies, such as excess inventory and storage costs, validating lean literature's emphasis on eliminating Muda. The prioritisation of Just-in-Time Production (0.3173) as the top solution highlights its dual strength in aligning with lean principles (C5) and mitigating implementation complexity (C4), a common barrier in practice. The Bayesian BWM's probabilistic outputs provided actionable confidence intervals, addressing AHP's limitation in quantifying uncertainty—e.g., Just-in-Time's superiority over Pull-Based Scheduling ($\Delta = 0.0613$) was statistically robust, represented using Figure 2. Our hybrid approach is designed to be both smart and practical. AHP has been a structured approach to make sure we were

identifying the correct problems and minimising bias. Then, Bayesian BWM helped us find reliable solutions, even when things get uncertain in the real world. This means practitioners can use our framework to prioritise the most effective interventions for their specific situation – like implementing Just-in-Time to match production with demand better and reduce lead times. The steps should be followed to are: Establish Real-Time Demand Forecasting, adopt a Pull-Based System, Collaborate with Reliable Suppliers, Reduce Lead Times, Design Cellular Manufacturing, Standardize Work Processes, Embed Quality at the Source, and Foster a Culture of Continuous Improvement.

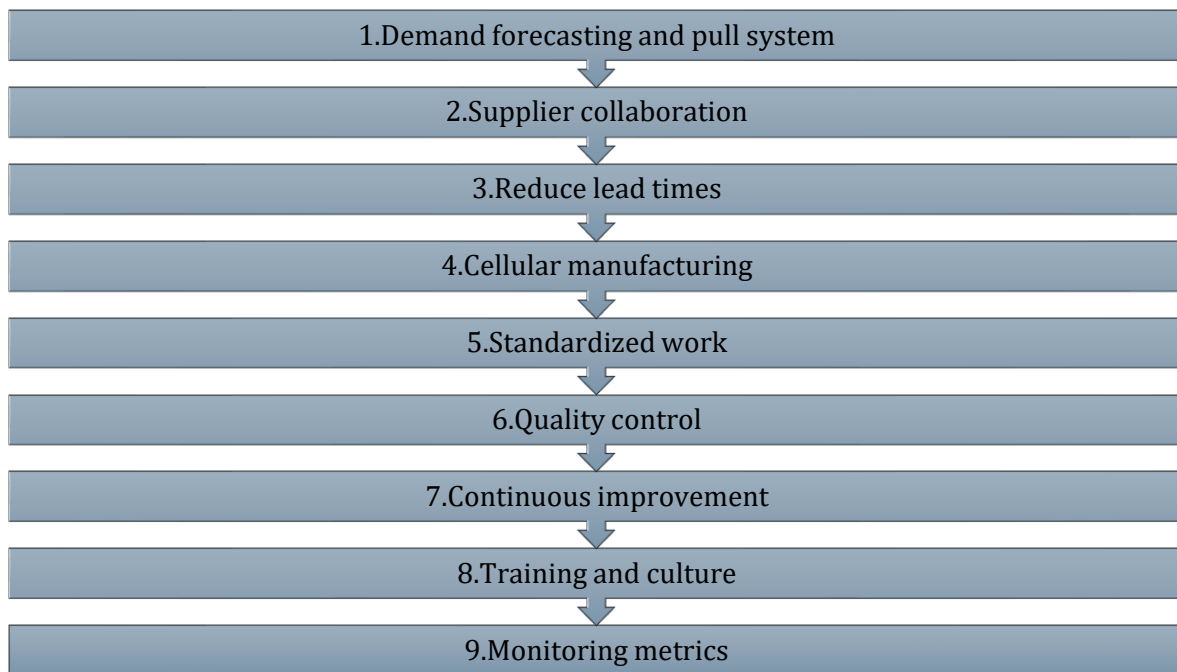


Figure 2. Steps in JIT (Just In Time) Production

5. Conclusion and Future Study:

The findings of this study underscore the critical importance of integrating structured decision-making frameworks like the Analytic Hierarchy Process (AHP) and Bayesian Best-Worst Method (BWM) to address the challenges of lean manufacturing (LM) systems. It has been confirmed that the dual-phase MCDM framework in this study is a successful method for determining the most significant issues in lean manufacturing and ranking context-specific fixes. Just-In-Time (JIT) manufacturing was shown to be the best way to lessen the effects of overproduction, which was found to be the most pressing problem. Using probabilistic reasoning, the hybrid AHP-Bayesian BWM strategy improved decision reliability under uncertainty while lowering subjective biases in problem identification. This framework gives businesses a useful, data-driven approach to methodically handle the problems associated with lean manufacturing, allowing them to maximize output, cut waste, and enhance efficiency. JIT production is ranked as the best option since it adheres to lean principles and can handle implementation complexity, which is a typical practice hurdle.

After the analysis and studies, it can be said that practising and maintaining lean in the industry is beneficial for success. Those integrated analyses gave us a crystal-clear idea and solution about lean manufacturing. Processes like Just in Time, Kanban System, and Kaizen should

be adequately implemented in the industry. In the future, we can compare our work process with another one to measure effectiveness. In the meantime, implementing those solutions at the industry level can be studied broadly. Although this study offers a thorough framework for tackling the difficulties associated with lean manufacturing, several directions for future research might increase its relevance and influence. Some examples are applications specific to a particular industry, integration with other MCDM techniques, real-time data integration, organizational and cultural aspects, sustainability integration, digital twins, and simulation.

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